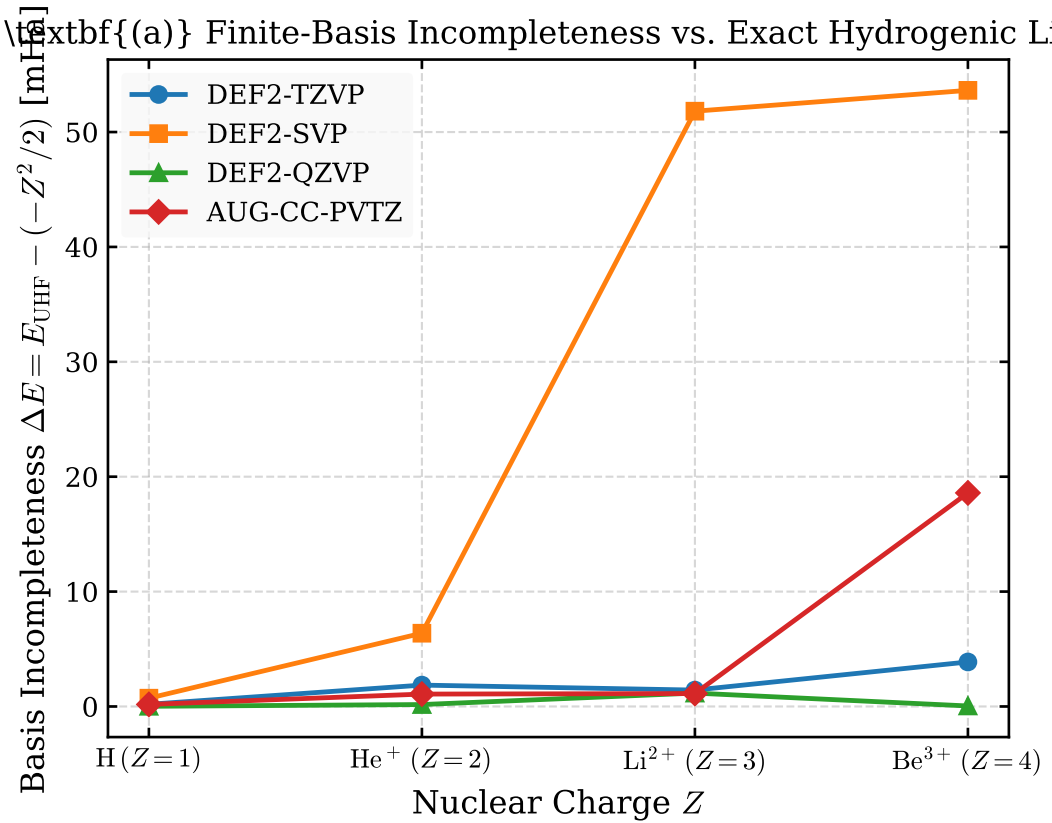
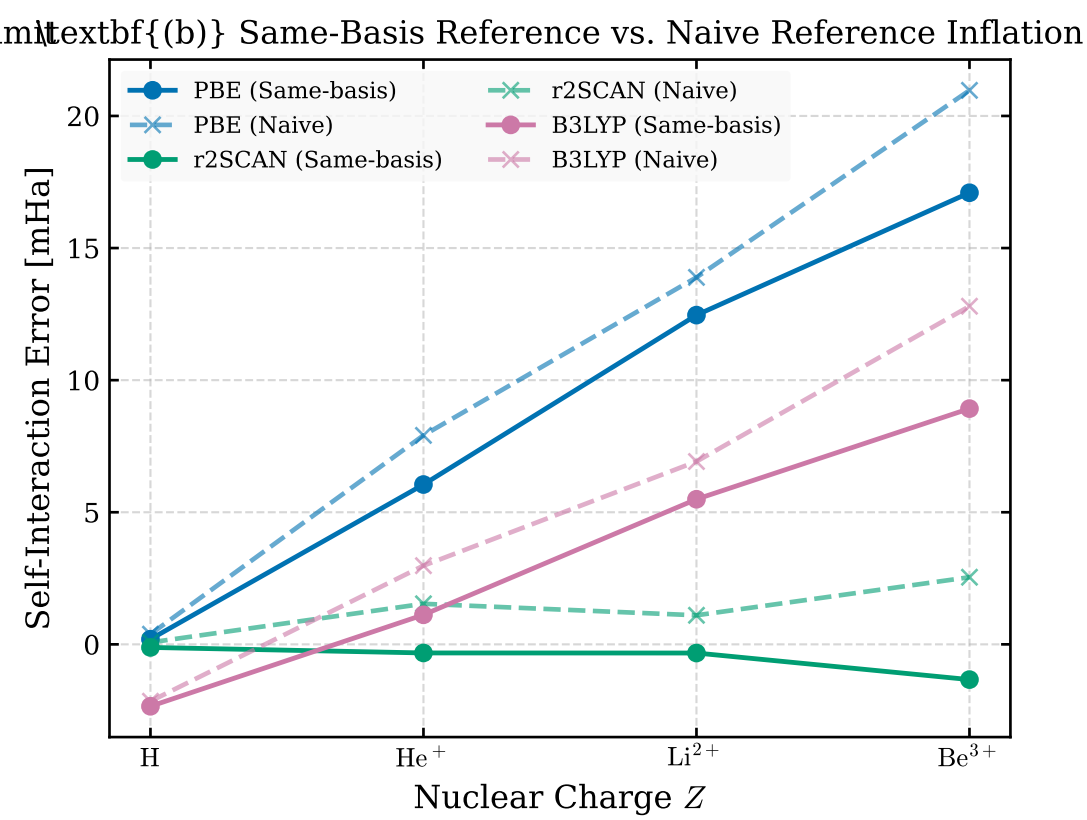

\textbf{(a)} Finite-Basis Incompleteness vs. Exact Hydrogenic Limit
Basis Incompleteness $\Delta E = E_{\mathrm{UHF}} - (-Z^2/2)$ [mHa]
DEF2-TZVP
DEF2-SVP
DEF2-QZVP
AUG-CC-PVTZ
0
10
20
30
40
50
H $(Z=1)$
He^+ $(Z=2)$
Li^{2+} $(Z=3)$
Be^{3+} $(Z=4)$
Nuclear Charge Z


\textbf{(b)} Same-Basis Reference vs. Naive Reference Inflation
Self-Interaction Error [mHa]
PBE (Same-basis)
PBE (Naive)
r2SCAN (Same-basis)
r2SCAN (Naive)
B3LYP (Same-basis)
B3LYP (Naive)
0
5
10
15
20
H
He^+
Li^{2+}
Be^{3+}
Nuclear Charge Z